

■ DATA STRUCTURES

Subject Code: R23 | Academic Year 2024-25 | 1st Year – 2nd Semester

Branch: CSE / IT & Allied

Credits: 3 (Theory) + Lab

Category: Core Subject

■ Course Objectives

- Understand basic & advanced data structures
- Learn efficient program design techniques
- Apply DS concepts in real-world problem solving

■ Course Outcomes

- Use arrays, linked lists, stacks & queues
- Solve problems using efficient algorithms
- Apply hashing & trees in real applications

UNIT – I : Introduction & Arrays

Introduction to Data Structures

- Definition and importance of Data Structures
- Types: Linear (Array, List, Stack, Queue) & Non-Linear (Tree, Graph)

Arrays

- Operations: Insertion, Deletion, Traversal
- Searching: Linear Search, Binary Search
- Sorting basics: Bubble, Selection, Insertion Sort

Time Complexity

- Big-O Notation – $O(1)$, $O(n)$, $O(n^2)$, $O(\log n)$
- Best, Average & Worst case analysis

UNIT – II : Linked Lists

Types of Linked Lists

- Singly Linked List – one directional traversal
- Doubly Linked List – forward & backward traversal
- Circular Linked List – last node points to head

Operations

- Insertion: at beginning, end, and middle
- Deletion: by value, by position
- Traversal & Searching

Memory Concepts

- Dynamic memory allocation using pointers
- malloc(), free() in C language

UNIT – III : Stacks & Queues

Stacks

- LIFO (Last In First Out) principle
- Operations: Push, Pop, Peek, isEmpty
- Applications: Recursion, Expression Evaluation (Infix→Postfix)
- Function call stack in memory

Queues

- Simple Queue – FIFO (First In First Out)
- Circular Queue – efficient memory utilization
- Priority Queue – elements served by priority
- Deque (Double-Ended Queue) – insert/delete at both ends

Applications

- CPU Scheduling, BFS Algorithm, Printer Spooling

UNIT – IV : Trees

Tree Terminology

- Root, Parent, Child, Leaf, Height, Depth, Degree
- Subtree, Level, Edge, Path

Binary Tree & BST

- Binary Tree – each node has at most 2 children
- Binary Search Tree (BST) – left < root < right
- Insertion, Deletion & Search in BST

Tree Traversals

- Inorder (Left → Root → Right) – gives sorted output in BST
- Preorder (Root → Left → Right) – used for tree copy
- Postorder (Left → Right → Root) – used for deletion

Heaps

- Min-Heap & Max-Heap concepts
- Heap property and heapify operation

UNIT – V : Hashing

Introduction to Hashing

- Hash Table – key-value pair storage
- Hash Function – maps key to table index
- Division method, Mid-square, Folding method

Collision Resolution

- Chaining – linked list at each slot
- Open Addressing: Linear Probing, Quadratic Probing, Double Hashing

Applications of Hashing

- Fast Searching – $O(1)$ average time
- Caching (Browser cache, DNS cache)
- Database Indexing & Symbol Tables in Compilers

■ Data Structures Lab (C Language)

- ◆ Array & Linked List programs in C
- ◆ Stack & Queue implementations
- ◆ Tree traversals (Inorder, Preorder, Postorder)
- ◆ Searching: Linear & Binary Search
- ◆ Sorting: Bubble, Selection, Insertion
- ◆ Hashing programs with collision handling

■ Important Notes & Tips

- ◆ Crucial subject for Placements & DSA rounds
- ◆ Focus on Coding – practice daily in C
- ◆ Master Time & Space Complexity analysis
- ◆ Understand logic before memorising syntax
- ◆ Solve at least 2–3 problems per topic